

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12413461
Kind Code	B2
Date of Patent	September 09, 2025
Inventor(s)	Harada; Hiroki et al.

Terminal and communication method

Abstract

A terminal includes a communication unit configured to transmit and receive a multi-carrier signal in a frequency band equal to or higher than a predetermined frequency, and a control unit configured to determine a first frame structure of the multi-carrier signal, wherein in the first frame structure, at least one of a subcarrier spacing and a number of resource blocks in a maximum channel bandwidth, a CP (cyclic prefix) ratio, and a time unit or a frequency unit of scheduling is different from a second frame structure of the multi-carrier signal in a frequency band equal to or lower than the predetermined frequency.

Inventors:	Harada; Hiroki (Tokyo, JP), Yoshioka; Shohei (Tokyo, JP), Kumagai; Shinya (Tokyo, JP), Kurita; Daisuke (Tokyo, JP), Nagata; Satoshi (Tokyo, JP)
Applicant:	NTT DOCOMO, INC. (Tokyo, JP)
Family ID:	77199829
Assignee:	NTT DOCOMO, INC. (Tokyo, JP)
Appl. No.:	17/793327
Filed (or PCT Filed):	February 07, 2020
PCT No.:	PCT/JP2020/005001
PCT Pub. No.:	WO2021/157094
PCT Pub. Date:	August 12, 2021

Prior Publication Data

Document Identifier	Publication Date
US 20230071680 A1	Mar. 09, 2023

Publication Classification

Int. Cl.: H04L27/26 (20060101); H04L5/00 (20060101); H04W72/12 (20230101)

U.S. Cl.:

CPC H04L27/2607 (20130101); H04L5/0094 (20130101); H04W72/12 (20130101);

Field of Classification Search

USPC: None

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
10476718	12/2018	Park	N/A	H04L 5/0048
2018/0091267	12/2017	Kim et al.	N/A	N/A
2019/0306832	12/2018	Si	N/A	H04L 5/0082
2019/0306923	12/2018	Xiong	N/A	H04L 27/261
2020/0067754	12/2019	Jung	N/A	H04L 5/0007
2020/0099476	12/2019	Park	N/A	H04W 4/70
2020/0099479	12/2019	Park	N/A	H04W 4/70
2020/0220669	12/2019	Park	N/A	H04L 1/1861
2020/0275412	12/2019	Kim	N/A	H04B 7/06954
2021/0136806	12/2020	Xiong	N/A	H04W 72/23
2021/0243624	12/2020	Moosavi	N/A	H04W 36/0061
2021/0281458	12/2020	Takeda	N/A	H04W 72/1268
2021/0297215	12/2020	Da	N/A	H04W 4/02
2021/0314785	12/2020	Raghavan	N/A	H04W 76/11
2021/0377943	12/2020	Park	N/A	H04L 5/001
2022/0166568	12/2021	Kim	N/A	H04L 5/0037
2022/0217513	12/2021	Back	N/A	H04W 92/10
2022/0271894	12/2021	Li	N/A	H04W 72/0446
2022/0345196	12/2021	Åström	N/A	H04B 7/0452
2022/0360353	12/2021	Pajukoski	N/A	H04L 5/0007
2022/0361209	12/2021	Zhu	N/A	H04W 72/1273
2022/0394626	12/2021	Matsumura	N/A	H04W 72/04
2023/0022908	12/2022	Yuan	N/A	H04L 5/0091
2023/0156543	12/2022	Xie	370/332	H04W 36/0088
2024/0305419	12/2023	Su	N/A	H04L 1/0061

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2018/062875	12/2017	WO	N/A

OTHER PUBLICATIONS

International Search Report issued in Application No. PCT/JP2020/005001 mailed on Aug. 4, 2020 (5 pages). cited by applicant

Written Opinion issued in Application No. PCT/JP2020/005001 mailed on Aug. 4, 2020 (4 pages). cited by applicant

3GPP TS 38.300 V15.8.0, “3rd Generation Partnership Project; Technical Specification Group Radio Access Network; NR; NR and NG-RAN Overall Description; Stage 2 (Release 15)” Dec. 2019, (55 pages). cited by applicant

NTT Docomo, Inc., “White Paper 5G Evolution and 6G” NTT Docomo, Inc., Jan. 2020 (17 pages). cited by applicant

3GPP TS 38.101-1 V15.8.2, “3rd Generation Partnership Project; Technical Specification Group Radio Access Network; NR; User Equipment (UE) radio transmission and reception; Part 1: Range 1 Standalone (Release 15)”, Dec. 2019 (5 pages). cited by applicant

3GPP TS 38.101-2 V15.8.0, “3rd Generation Partnership Project; Technical Specification Group Radio Access Network; NR; User Equipment (UE) radio transmission and reception; Part 2: Range 2 Standalone (Release 15)”, Dec. 2019 (37 pages). cited by applicant

3GPP TS 38.101-3 V16.2.1, “3rd Generation Partnership Project; Technical Specification Group Radio Access Network; NR; User Equipment (UE) radio transmission and reception; Part 3: Range 1 and Range 2 Interworking operation with other radios (Release 16)”, Dec. 2019 (19 pages). cited by applicant

Office Action issued in counterpart Chinese Patent Application No. 202080094216.2 mailed on Nov. 18, 2024 (16 pages). cited by applicant

Office Action issued in Chinese Patent Application No. 202080094216.2, mailed Jun. 18, 2025 (12 pages). cited by applicant

Primary Examiner: Phunkulh; Bob A

Attorney, Agent or Firm: Osha Bergman Watanabe & Burton LLP

Background/Summary

TECHNICAL FIELD

(1) The present invention relates to a terminal and a communication method in a wireless communication system.

BACKGROUND ART

(2) In the NR (New Radio) (also referred to as “5G”), which is the successor system of LTE (Long Term Evolution), techniques for satisfying, as required conditions, large capacity system, high data transmission speed, low delay, and simultaneous connection of many terminals, low cost, power saving, and the like are being studied (for example, Non-Patent Document 1). 5G is a mobile communication system that supports higher frequency bands such as millimeter waves of more than 10 GHz. Ultra-high-speed wireless data communication in the class of several Gbps can be implemented by using a frequency bandwidth of several hundred MHz, which is significantly wider than conventional systems such as LTE.

(3) However, scope for improvement is associated with millimeter-wave technology in mobile communication, and discussion of Beyond 5G and 6G as post-5G systems have begun (for example, Non-Patent Document 2). Expansion of new frequency bands is being discussed as a candidate discussion item for

(4) Beyond 5G and 6G. For example, the new frequency bands include a frequency band of 100 GHz or higher, a terahertz band, or the like.

PRIOR ART DOCUMENT

Non-Patent Document

(5) Non-Patent Document 1: 3GPP TS 38.300 V15.8.0 (2019-12) Non-Patent Document 2: NTT DOCOMO, INC., DOCOMO 5G White Paper “5G Evolution and 6G” (2020-01) Non-Patent Document 3: 3GPP TS 38.101-1 V15.8.2 (2019-12) Non-Patent Document 4: 3GPP TS 38.101-2 V15.8.0 (2019-12)

SUMMARY OF THE INVENTION

Problem to be Solved by the Invention

(6) However, in a higher frequency band, in a case where a multi-carrier signal such as an OFDM (Orthogonal Frequency Division Multiplexing) is used, for example, there are problems such as non-linearity of an amplifier, phase noises, PAPR (Peak-to-Average Power Ratio) increasing when the number of subcarriers is large, and the like.

(7) The present invention has been made in view of the above problem, and it is an object of the present invention to perform communication using a multi-carrier signal suitable for the higher frequency band in a wireless communication system.

Means for Solving Problem

(8) According to the disclosed technique, provided is a terminal including a communication unit configured to transmit and receive a multi-carrier signal in a frequency band equal to or higher than a predetermined frequency, and a control unit configured to determine a first frame structure of the multi-carrier signal, wherein in the first frame structure, at least one of a subcarrier spacing and a number of resource blocks in a maximum channel bandwidth, a CP (cyclic prefix) ratio, and a time unit or a frequency unit of scheduling is different from a second frame structure of the multi-carrier signal in a frequency band equal to or lower than the predetermined frequency.

Effect of the Invention

(9) According to the disclosed technique, communication can be performed using a multi-carrier signal suitable for the higher frequency band in a wireless communication system.

Description

BRIEF DESCRIPTION OF DRAWINGS

- (1) FIG. 1 is a drawing illustrating a configuration example (1) according to an embodiment of the present invention wireless communication system;
- (2) FIG. 2 is a drawing illustrating a configuration example (2) according to the embodiment of the present invention wireless communication system;
- (3) FIG. 3 is a drawing illustrating an example of frequency bands according to the embodiment of the present invention;
- (4) FIG. 4 is a drawing for explaining an OFDM signal and a single-carrier signal;
- (5) FIG. 5 is a drawing for explaining a nested structure;
- (6) FIG. 6 is a drawing for explaining an example of a frame structure according to the embodiment of the present invention;
- (7) FIG. 7 is a drawing illustrating an example (1) of CP insertion according to the embodiment of the present invention;
- (8) FIG. 8 is a drawing illustrating an example (2) of CP insertion according to the embodiment of the present invention;
- (9) FIG. 9 is a drawing illustrating an example of a functional configuration of a base station **10** according to the embodiment of the present invention;
- (10) FIG. **10** is a drawing illustrating an example of a functional configuration of a terminal **20** according to the embodiment of the present invention; and
- (11) FIG. **11** is a drawing illustrating an example of a hardware configuration of the base station **10** or the terminal **20** according to the embodiment of the present invention.

MODE FOR CARRYING OUT THE INVENTION

(12) An embodiment of the present invention will be hereinafter described with reference to drawings. The embodiment described below is an example, and the embodiment to which the present invention is applied is not limited to the following embodiment.

(13) In operation of a wireless communication system according to embodiments of the present invention, existing techniques are used as appropriate. However, an example of existing technique includes an existing LTE, but is not limited to the existing LTE. In addition, the term “LTE” used in this specification has a broad meaning including LTE-Advanced and specifications newer than LTE-Advanced (e.g., NR) unless otherwise specified.

(14) In the embodiments of the present invention described below, terms such as SS (Synchronization signal), PSS (Primary SS), SSS (Secondary SS), PBCH (Physical broadcast channel), PRACH (Physical random access channel), PDCCH (Physical Downlink Control Channel), PDSCH (Physical Downlink Shared Channel), PUCCH (Physical Uplink Control Channel), PUSCH (Physical Uplink Shared Channel), and the like used in the existing LTE are used. This is for convenience of description, and signals, functions, and the like may be referred to as other names. In the NR, the above terms correspond to NR-SS, NR-PSS, NR-SSS, NR-PBCH, NR-PRACH, NR-PDCCH, NR-PDSCH, NR-PUCCH, NR-PUSCH, and the like. However, even when signals are used for NR, “NR-” is not necessarily attached thereto.

(15) In the embodiments of the present invention, the duplex method may be a TDD (Time Division Duplex) system, an FDD (Frequency Division Duplex) system, or others (for example, Flexible Duplex and the like).

(16) Further, in the embodiment of the present invention, “to configure” or “to specify” a radio parameter or the like may be that a predetermined value is configured in advance (pre-configured) in the base station **10** or the terminal **20**, may be that a predetermined value is assumed to be configured in advance (pre-configured) in the base station **10** or the terminal **20**, or may be that a radio parameter notified from the base station **10** or the terminal **20** is configured.

(17) FIG. 1 is a drawing illustrating a configuration example (1) of a wireless communication system according to the embodiment of the present invention. As illustrated in FIG. 1, a base station **10** and a terminal **20** are included. In FIG. 1, one base station **10** and one terminal **20** are illustrated, but this is only an example. Alternatively, a plurality of base stations **10** and terminals **20** may be provided. The terminal **20** may also be referred to as “user equipment”. The wireless communication system according to the present embodiment may be referred to as a NR-U system.

(18) The base station **10** provides one or more cells, and is a communication apparatus wirelessly communicating with the terminal **20**. The physical resource of a radio signal may be defined by the time domain and the frequency domain. The time domain may be defined by slots or an OFDM symbols. The frequency domain may be defined by sub-bands, subcarriers, or resource blocks.

(19) As illustrated in FIG. 1, the base station **10** transmits control information or data to the terminal **20** through DL (Downlink), and receives control information or data from the terminal **20** through UL (Uplink). Both the base station **10** and the terminal **20** can transmit and receive signals by performing beamforming. Both of the base station **10** and the terminal **20** can apply communication based on MIMO (Multiple Input Multiple Output) to DL or UL. Also, both of the base station **10** and the terminal **20** may perform communication via a secondary cell (SCell) with CA (Carrier Aggregation) and a primary cell (PCell).

(20) The terminal **20** is a communication apparatus equipped with a wireless communication function such as a smartphone, a mobile phone, a tablet, a wearable terminal, and a communication module for M2M (Machine-to-Machine). As illustrated in FIG. 1, the terminal **20** receives control information or data from the base station **10** in DL, and transmits control information or data to the base station **10** in UL, thereby using various communication services provided by the wireless communication system.

(21) FIG. 2 is a drawing illustrating a configuration example (2) of a wireless communication system according to the embodiment of the present invention. FIG. 2 illustrates an example of

configuration of the wireless communication system in a case where NR-DC (NR-Dual connectivity) is executed. As illustrated in FIG. 2, a base station **10A** serving as a MN (Master Node) and a base station **10B** serving as a SN (Secondary Node) are provided. The base station **10A** and the base station **10B** are connected to a core network **30**. The terminal **20** communicates with both of the base station **10A** and the base station **10B**.

(22) A cell group provided by the base station **10A**, i.e., the MN, is referred to as a MCG (Master Cell Group), and a cell group provided by the base station **10B**, i.e., the SN, is referred to as a SCG (Secondary Cell Group). The following operation may be performed by any of the configurations of FIG. 1 and FIG. 2. The SCG may include a PSCell (Primary SCG Cell) and an SCell. The PCell may be a PCell of the MCG, or may be a PSCell of the SCG.

(23) Discussions of Beyond 5G and 6G as post-5G systems have begun. Expansion of new frequency bands is being discussed as a candidate discussion item for Beyond 5G and 6G. For example, the new frequency bands include a frequency band of 100 GHz or higher, a terahertz band, or the like.

(24) FIG. 3 is a drawing illustrating an example of frequency bands according to the embodiment of the present invention. As illustrated in FIG. 3, the FR1 is defined as a frequency band up to 7.125 GHz with subcarrier spacings {15, 30, 60} kHz and with a band width of 5 to 100 MHz. As illustrated in FIG. 3, the FR2 is defined as a frequency band of 24.25 to 52.6 GHz with subcarrier spacings {60, 120, 240} kHz and with a bandwidth of 50 to 400 MHz. As illustrated in FIG. 3, a frequency band of 52.6 to 71 GHz may be referred to as "FR2x".

(25) The frequency band of 52.6 to 71 GHz illustrated in FIG. 3 is discussed to be supported in NR Release 17. In the NR Release 16, the frequency usage, use cases, and required specifications of 52.6 to 114.25 GHz in each country has been discussed, and therefore, in NR Release 18 and later, support may be extended to around 114 GHz.

(26) However, for example, in a higher frequency band and the like of 100 GHz or higher, due to problems such as non-linearity of an amplifier, phase noises, and the like, a single-carrier signal is more suitable than a multi-carrier signal such as OFDM (Orthogonal Frequency Division Multiplexing).

(27) For example, as the number of subcarriers increases, a ratio of peak power to average power, PAPR (Peak-to-Average Power Ratio), with the matched phases increases in the OFDM signal, which makes it difficult to increase the transmission power. Also, in the higher frequency band, signals are less likely to propagate through multipath propagation due to a large reflection loss and propagation loss. There is a possibility that the advantages of CP (cyclic prefix)-OFDM, which is advantageous in multipath propagation with a high affinity with MIMO, are not made use of effectively.

(28) On the other hand, when a wide band, i.e., several GHz, is used with a single-carrier waveform, there is a problem in that the frequency utilization efficiency decreases. FIG. 4 is a drawing for explaining an OFDM signal and a single-carrier signal. When an OFDM signal and a single-carrier signal illustrated in FIG. 4 are compared, a signal in which carrier spacings or subcarrier spacings are tightly arranged without interference is the OFDM signal, and theoretically, the frequency utilization efficiency over the entire band where data is transmitted in parallel with multiple carriers by using the OFDM signal, is superior to the single-carrier signal.

(29) When carrier aggregation and the like with existing frequency bands are considered, the symbol length or the slot length is preferably in a nested structure. The nested structure means, for example, a structure in which a single symbol length or slot length in a lower frequency band is the same as a symbol length or a slot length of $2^{\text{sup.n}}$ (power of two) symbols or slots in a high frequency band. FIG. 5 is a drawing for explaining the nested structure. As illustrated in FIG. 5, for example, 14 symbols are arranged in a single slot in the lowest frequency band. In the second lowest frequency band, two slots, i.e., 28 symbols, are arranged in the period corresponding to a single slot length of the lowest frequency band, and the second lowest frequency band includes

symbols and slots twice as many as the symbols and the slots in a predetermined period of the lowest frequency band. In the highest frequency band, 4 slots, i.e., 56 symbols, are arranged in the period corresponding to a single slot length of the lowest frequency band, and the highest frequency band includes symbols and slots four times as many as the symbols and the slots in a predetermined period of the lowest frequency band. The frame structure as illustrated in FIG. 5 is an example of nested structure, and in this case, for example, when scheduling or feedback is performed with cross carrier, the timing can be determined easily.

(30) In the case of the single-carrier signal, a symbol rate is determined by an occupied bandwidth, and therefore, it may be difficult to set the symbol length to an appropriate length while using a wide band. In contrast, in the case of the OFDM signal, the symbol length is determined by subcarrier spacings, and the occupied bandwidth can be adjusted flexibly to some extent by the number of subcarriers. Accordingly, cross carrier processing can be performed easily with a relationship between the symbol length and the slot length with the lower frequency band being a nested structure while occupying the available bandwidth as much as possible.

(31) Therefore, a channel structure and a frame structure are specified in view of the efficiency in the use of an OFDM signal in the higher frequency band and in view of an affinity with a frame structure for the lower frequency band.

(32) For example, in a case where the OFDM is applied to a predetermined frequency or higher, it may be specified as shown in the following A) to C). The predetermined frequency may be, for example, a frequency higher than the FR2, such as 52.6 GHz, 71 GHz, 114.25 GHz, or the like.

(33) A) The upper limit of the number of subcarriers or/and the upper limit of the number of resource blocks per available carrier (CC) at the predetermined frequency or higher may be specified as a value lower than an available value at the predetermined frequency or lower. In other words, the terminal **20** and the base station **10** may configure, as a value lower than the available value at the predetermined frequency or lower, the upper limit of the number of subcarriers or/and the upper limit of the number of resource blocks per available carrier (CC) at the predetermined frequency or higher. For example, the terminal **20** and the base station **10** may assume, at the predetermined frequency or higher, the number of points of FFT (Fast Fourier Transform) that is less than the number of points of FFT assumed at the predetermined frequency or lower.

(34) B) A ratio of a CP length to a symbol length (CP ratio) at the predetermined frequency or higher may be specified as a ratio lower than a ratio of a CP length to a symbol length used at the predetermined frequency or lower, or it may be specified that a CP is not inserted at the predetermined frequency or higher. In other words, the terminal **20** and the base station **10** may configure, as a ratio lower than the ratio of the CP length to the symbol length used at the predetermined frequency or lower, the ratio of the CP length to the symbol length (CP ratio) at the predetermined frequency or higher, or may configure a CP not being inserted at the predetermined frequency or higher. For example, a length in which an integer number N of symbols including CPs are arranged, may match at least a length in which an integer number M (M is less than N) of symbols including CPs used at the predetermined frequency or lower are arranged. In other words, during CA, the slot border of some of at least some of the slots may match between carriers.

(35) C) At least one of: the number of symbols constituting a time unit (for example, a slot) serving as a unit of scheduling; or the number of subcarriers constituting a frequency unit (for example, a resource block) at the predetermined frequency or higher, may be specified as a value larger than a value at the predetermined frequency or lower. In other words, the terminal **20** and the base station **10** may configure, as a value larger than the value at the predetermined frequency or lower, at least one of: the number of symbols constituting a time unit (for example, a slot) serving as a unit of scheduling; or the number of subcarriers constituting a frequency unit (for example, a resource block) at the predetermined frequency or higher. Alternatively, at least one of the number of symbols or the number of subcarriers, serving as the unit of scheduling, may be configured to be indicated by the base station **10** to the terminal **20** on the basis of the configuration.

20 and the base station **10** may insert a CP only before the first symbol. When the CP is inserted only before the first symbol, a ratio of the CP to one symbol length may be larger than the lower frequency band.

(50) FIG. **6** is a drawing for explaining an example of a frame structure according to the embodiment of the present invention. As illustrated in FIG. **6**, in a time length of 1 slot, i.e., 14 symbols, including CPs in the lower frequency band, where slots and symbols scheduled while the CP ratio according to the subcarrier spacing is maintained are arranged, 4 slots, i.e., 56 symbols, are arranged in the same time length as long as the subcarrier spacing is a multiple of a predetermined time range. In this case, the above specification of the CP in the higher frequency band is applied, and the number of symbols arranged in the same time length can be increased by removing the CP or minimizing the CP. For example, as illustrated in FIG. **16**, in a case where the subcarrier spacing is four times the subcarrier spacing of the lower frequency band similarly with the above example, 3 symbols may be increased by using the time obtained by minimizing the CP length, so that 59 symbols from symbol #0 to symbol #58 can be arranged in the same time length.

(51) FIG. **7** is a drawing illustrating an example (1) of CP insertion according to the embodiment of the present invention. As illustrated in FIG. **7**, the terminal **20** and the base station **10** may insert the minimized CP before each symbol. The length of the CP inserted before each symbol may be different for each symbol. For example, the length of a CP inserted before the first symbol may be longer than the length of a CP inserted before any one of the symbols.

(52) FIG. **8** is a drawing illustrating an example (2) of CP insertion according to the embodiment of the present invention. As illustrated in FIG. **8**, the terminal **20** and the base station **10** may insert a minimized CP before an aggregate of multiple symbols. The length of a CP inserted before an aggregate of multiple symbols may be different for each symbol. For example, the length of a CP inserted before multiple successive symbols to which a reference signal is mapped may be longer than the length of a CP inserted before any other multiple symbols.

(53) Hereinafter, a case where the above C) is specified is described in detail. The NR employs a frame structure in which, regardless of the subcarrier spacing or the frequency band, one slot includes 14 symbols, and one resource block includes 12 subcarriers.

(54) Therefore, at the predetermined frequency (for example, 52.6 GHz, 71 GHz, 114.25 GHz, or the like) or higher, the terminal **20** and the base station **10** may set, to a value larger than 14, the number of symbols constituting a time unit of scheduling.

(55) For example, the terminal **20** and the base station **10** causes the length of an integer number of units (for example, slots) in which more symbols are arranged due to the minimized CP length to match the length of an integer multiple of a slot size in the lower frequency band, so that specifications for slot position designation during cross carrier scheduling or cross carrier triggering or slot timing designation during feedback can be simplified.

(56) In addition, the terminal **20** and the base station **10** may cause the number of subcarriers constituting a frequency unit of scheduling to be a value larger than 12. For example, in an environment where there is almost no multipath, there is no frequency selectivity in the loss, and accordingly, there is no need to increase the granularity of scheduling in the frequency domain, and by causing the number of subcarriers constituting a frequency unit of scheduling to be a value larger than 12, the terminal **20** and the base station **10** can reduce a notification bit size associated with the frequency resource.

(57) The size of a unit in the time domain for scheduling and the size of a unit in the frequency domain for scheduling may be configurable. According to the environment, the scenario of installation of base stations, and the like, the network can determine and configure the CP length or the scheduling granularity. In other words, the base station **10** can configure, as radio parameters, the CP length or the scheduling granularity for the terminal **20**. For example, in an environment where there is almost no multipath in the higher frequency band, the delay spread is small, and therefore, the base station **10** may determine the CP length as a small value, and configure the

determined CP length for the terminal **20**. In an environment where there is almost no multipath in the higher frequency band, the frequency selectivity is weak, and therefore, the base station **10** may determine the scheduling granularity as a large value, and configure the determined scheduling granularity for the terminal **20**.

(58) According to the above embodiment, in a case where the OFDM is applied in the predetermined frequency or higher, the terminal **20** can improve the efficiency of communication by adopting a flexibly configurable frame structure suitable for the environment of the higher frequency band.

(59) In other words, communication can be performed using a multi-carrier signal suitable for the higher frequency band in a wireless communication system.

Apparatus Configuration

(60) Next, an example of functional configuration of the base station **10** and the terminal **20** that execute the processing and operations described so far will be described. The base station **10** and the terminal **20** include a function for implementing the above-described embodiment. However, each of the base station **10** and the terminal **20** may have only some of the functions in the embodiment.

Base Station **10**

(61) FIG. **9** is a drawing illustrating an example of a functional configuration of the base station **10**. As illustrated in FIG. **9**, the base station **10** includes a transmitting unit **110**, a receiving unit **120**, a configuring unit **130**, and a control unit **140**. The functional configuration illustrated in FIG. **9** is only an example. As long as the operation according to the embodiment of the present invention can be executed, the functions may be divided in any way, and the functional units may be given any names.

(62) The transmitting unit **110** includes a function of generating signals to be transmitted to the terminal **20** and wirelessly transmitting the signals. Also, the transmitting unit **110** transmits an inter-network node message to another network node. The receiving unit **120** includes a function of wirelessly receiving various types of signals transmitted from the terminal **20** and acquiring, for example, information on a higher layer from the received signals. Further, the transmitting unit **110** has a function of transmitting NR-PSS, NR-SSS, NR-PBCH, a DL/UL control signal, a reference signal or the like to the terminal **20**. Also, the receiving unit **120** receives an inter-network node message from another network node. The transmitting unit **110** and the receiving unit **120** may be combined into a communication unit.

(63) The configuring unit **130** stores configuration information configured in advance and various configuration information to be transmitted to the terminal **20** in a storage device and reads out the configuration information from the storage device as needed. The contents of the configuration information include, for example, information used for cross carrier scheduling.

(64) As described in the embodiment, the control unit **140** performs control of cross carrier scheduling. Also, the control unit **140** performs control of a HARQ response. A functional unit configured to transmit signals in the control unit **140** may be included in the transmitting unit **110**, and a functional unit configured to receive signals in the control unit **140** may be included in the receiving unit **120**.

(65) <Terminal **20**>

(66) FIG. **10** is a drawing illustrating an example of a functional configuration of the terminal **20** according to the embodiment of the present invention. As illustrated in FIG. **10**, the terminal **20** includes a transmitting unit **210**, a receiving unit **220**, a configuring unit **230**, and a control unit **240**. The functional configuration illustrated in FIG. **10** is merely an example. As long as the operation according to the embodiment of the present invention can be executed, the functions may be divided in any way, and the function units may be given any names.

(67) The transmitting unit **210** has a function of generating a transmission signal from transmission data and wirelessly transmitting the transmission signal. The receiving unit **220** wirelessly receives

various types of signals, and acquires a signal in a higher-layer from the received signal in the physical layer. Also, the receiving unit **220** has a function of receiving NR-PSS, NR-SSS, NR-PBCH, DL/UL/SL control signals, reference signals, and the like that are transmitted from the base station **10**. Also, for example, in D2D communication, the transmitting unit **210** transmits, to another terminal **20**, a PSCCH (Physical Sidelink Control Channel), a PSSCH (Physical Sidelink Shared Channel), a PSDCH (Physical Sidelink Discovery Channel), a PSBCH (Physical Sidelink Broadcast Channel), and the like. The receiving unit **220** receives the PSCCH, the PSSCH, the PSDCH, the PSBCH, and the like, from the another terminal **20**. The transmitting unit **210** and the receiving unit **220** may be combined into a communication unit.

(68) The configuring unit **230** stores various types of configuration information received from the base station **10** or a terminal **20** by the receiving unit **220** in a storage device and reads out the configuration information from the storage device as needed. The configuring unit **230** also stores configuration information configured in advance. The contents of the configuration information include, for example, information used for cross carrier scheduling.

(69) As described in the embodiment, the control unit **240** performs control of cross carrier scheduling. The control unit **240** executes control of a HARQ response. A functional unit configured to transmit signals in the control unit **240** may be included in the transmitting unit **210**, and a functional unit configured to receive signals in the control unit **240** may be included in the receiving unit **220**.

(70) <Hardware Configuration>

(71) The block diagrams (FIGS. **9** and **10**) used for explaining the above embodiments illustrate blocks in units of functions. These functional blocks (constituting units) are implemented by any combinations of at least one of hardware and software. In this regard, a method for implementing the various functional blocks is not particularly limited. That is, each functional block may be implemented by one device united physically and logically. Alternatively, each functional block may be implemented by connecting directly or indirectly (for example, in a wired or wireless manner) two or more devices that are physically or logically separated and connected together and using these multiple devices. The functional block may be implemented by combining software with the single device or multiple devices.

(72) Functions include, but are not limited to, determining, calculating, processing, deriving, investigating, searching, confirming, receiving, transmitting, outputting, accessing, resolving, selecting, establishing, comparing, assuming, expecting, considering, broadcasting, notifying, communicating, forwarding, configuring, reconfiguring, allocating, mapping, assigning, and the like. For example, a functional block (constituting unit) that has a function of transmitting is referred to as a transmitting unit or a transmitter. As described above, a method for implementing these functions is not particularly limited.

(73) For example, the base station **10**, the terminal **20**, and the like according to one embodiment of the present disclosure may function as a computer that performs processing of a wireless communication according to the present disclosure. FIG. **11** is a drawing illustrating an example of a hardware configuration of the base station **10** or the terminal **20** according to an embodiment of the present disclosure. Each of the base station **10** and terminal **20** may be physically configured as a computer device including a processor **1001**, a storage device **1002**, an auxiliary storage device **1003**, a communication device **1004**, an input device **1005**, an output device **1006**, a bus **1007**, and the like.

(74) It is noted that, in the following description, the term “device” may be read as a circuit, an apparatus, a unit, or the like. The hardware configurations of the base station **10** and the terminal **20** may be configured to include one or more of the devices illustrated in drawings, or may be configured not to include some of the devices.

(75) Each function of the base station **10** and the terminal **20** may be implemented by reading predetermined software (program) to hardware such as the processor **1001**, the storage device

1002, or the like, causing the processor **1001** to perform operations, controlling communication by the communication device **1004**, and controlling at least one of reading and writing of data in the storage device **1002** and the auxiliary storage device **1003**.

(76) The processor **1001** executes, for example, an operating system to control the overall operation of the computer. The processor **1001** may be a central processing unit (CPU) including an interface with peripheral devices, a control device, an arithmetic device, a register, and the like. For example, the control unit **140**, the control unit **240**, and the like described above may be realized by the processor **1001**.

(77) The processor **1001** reads a program (program code), a software module, or data from at least one of the auxiliary storage device **1003** and the communication device **1004** onto the storage device **1002**, and performs various processes according to the program, the software module, or the data. As the program, a program that causes a computer to perform at least some of the operations described in the embodiment explained above is used. For example, the control unit **140** of the base station **10**, as illustrated in FIG. **9**, may be implemented by a control program that is stored in the storage device **1002** and that is executed by the processor **1001**. Also, for example, the control unit **240** of the terminal **20**, as illustrated in FIG. **10**, may be implemented by a control program that is stored in the storage device **1002** and that is executed by the processor **1001**. Explanation has been provided above for the case in which the above various processing are performed by the single processor **1001**. However, such processing may be simultaneously or sequentially performed by two or more processors **1001**. The processor **1001** may be implemented with one or more chips. It is noted that the program may be transmitted from a network through an electronic communication line.

(78) The storage device **1002** is a computer-readable recording medium and may be constituted by at least one of, for example, a ROM (Read Only Memory), an EPROM (Erasable Programmable ROM), an EEPROM (Electrically Erasable Programmable ROM), a RAM (Random Access Memory), and the like. The storage device **1002** may also be referred to as a register, a cache, a main memory (main storage device), or the like. The storage device **1002** can store a program (program code), a software module and the like that can be executed to perform a communication method according to an embodiment of the present disclosure.

(79) The auxiliary storage device **1003** is a computer-readable recording medium and may be configured by at least one of, for example, an optical disk such as a CD-ROM (Compact Disc ROM), a hard disk drive, a flexible disk, a magneto-optical disk (for example, a compact disk, a digital versatile disk, or a Blu-ray (registered trademark) disk), a smart card, a flash memory (for example, a card, a stick, or a key drive), a floppy (registered trademark) disk, a magnetic strip, and the like. The above storage medium may be, for example, a database, a server, or other appropriate media including at least one of the storage device **1002** and the auxiliary storage device **1003**.

(80) The communication device **1004** is hardware (a transmission and reception device) for performing communication between computers through at least one of a wired and wireless networks and may also be referred to as, for example, a network device, a network controller, a network card, a communication module, or the like. The communication device **1004** may include, for example, a radio frequency switch, a duplexer, a filter, a frequency synthesizer, or the like to implement at least one of a frequency division duplex (FDD) and a time division duplex (TDD). For example, a transmission and reception antenna, an amplifier, a transmitting and receiving unit, a transmission line interface, and the like may be implemented by the communication device **1004**. The transmitting and receiving unit may be implemented in such a manner that a transmitting unit and a receiving unit are physically or logically separated.

(81) The input device **1005** is an input device (for example, a keyboard, a mouse, a microphone, a switch, a button, a sensor, or the like) that receives an input from the outside. The output device **1006** is an output device (for example, a display, a speaker, an LED lamp, or the like) that performs an output to the outside. It is noted that the input device **1005** and the output device **1006** may be

integrated with each other (for example, a touch panel).

(82) The devices, such as the processor **1001** and the storage device **1002**, are connected to each other via a bus **1007** for communicating information. The bus **1007** may be constituted by using a single bus, or may be constituted by using busses different depending on devices.

(83) The base station **10** and the terminal **20** may include hardware, such as a microprocessor, a digital signal processor (DSP), an ASIC (Application Specific Integrated Circuit), a PLD (Programmable Logic Device), or an FPGA (Field Programmable Gate Array), or alternatively, some or all of the functional blocks may be implemented by the hardware. For example, the processor **1001** may be implemented with at least one of these hardware components.

Summary of Embodiment

(84) As described above, according to the embodiment of the present invention, provided is a terminal including a communication unit configured to transmit and receive a multi-carrier signal in a frequency band equal to or higher than a predetermined frequency, and a control unit configured to determine a first frame structure of the multi-carrier signal, wherein in the first frame structure, at least one of a subcarrier spacing and a number of resource blocks in a maximum channel bandwidth, a CP (cyclic prefix) ratio, and a time unit or a frequency unit of scheduling is different from a second frame structure of the multi-carrier signal in a frequency band equal to or lower than the predetermined frequency.

(85) According to the above structure, in a case where the OFDM is applied in the predetermined frequency or higher, the terminal **20** can improve the efficiency of communication by adopting a flexibly configurable frame structure suitable for the environment of the higher frequency band. In other words, communication can be performed using a multi-carrier signal suitable for the higher frequency band in a wireless communication system.

(86) In the first frame structure, the subcarrier spacing may be larger than that in the second frame structure, and the number of resource blocks in the maximum channel bandwidth may be smaller than that in the second frame structure. According to this configuration, in a case where the OFDM is applied in the predetermined frequency or higher, the terminal **20** can improve the efficiency of communication by reducing the number of points of FFT.

(87) In the first frame structure, a ratio of a CP length to a symbol length may be smaller than that in the second frame structure. According to this configuration, in a case where the OFDM is applied in the predetermined frequency or higher, the terminal **20** can improve the efficiency of communication by shortening the redundant CP length.

(88) The first frame structure may have a same subcarrier spacing as the second frame structure, and may include more symbols in a given time length than the second frame structure. According to this configuration, in a case where the OFDM is applied in the predetermined frequency or higher, the terminal **20** can improve the efficiency of communication by using more resources by shortening the redundant CP length.

(89) In the first frame structure, the time unit or the frequency unit of the scheduling may be larger than that in the second frame structure. According to this configuration, in a case where the OFDM is applied in the predetermined frequency or higher, the terminal **20** can improve the efficiency of communication by reducing the bit size for scheduling notification without increasing less needed granularity.

(90) Also, according to the embodiment of the present invention, provided is a communication method causing a terminal to execute transmitting and receiving a multi-carrier signal in a frequency band equal to or higher than a predetermined frequency, and determining a first frame structure of the multi-carrier signal, wherein in the first frame structure, at least one of a subcarrier spacing and a number of resource blocks in a maximum channel bandwidth, a CP (cyclic prefix) ratio, and a time unit or a frequency unit of scheduling is different from a second frame structure of the multi-carrier signal in a frequency band equal to or lower than the predetermined frequency.

(91) According to the above structure, in a case where the OFDM is applied in the predetermined

frequency or higher, the terminal **20** can improve the efficiency of communication by adopting a flexibly configurable frame structure suitable for the environment of the higher frequency band. In other words, communication can be performed using a multi-carrier signal suitable for the higher frequency band in a wireless communication system.

Supplements to Embodiment

(92) The embodiment of the present invention has been described above, but the disclosed invention is not limited to the above embodiment, and those skilled in the art would understand that various modified examples, revised examples, alternative examples, substitution examples, and the like can be made. In order to facilitate understanding of the present invention, specific numerical value examples are used for explanation, but the numerical values are merely examples, and any suitable values may be used unless otherwise stated. Classifications of items in the above description are not essential to the present invention, contents described in two or more items may be used in combination if necessary, and contents described in an item may be applied to contents described in another item (unless a contradiction arises). The boundaries between the functional units or the processing units in the functional block diagrams do not necessarily correspond to the boundaries of physical components. Operations of a plurality of functional units may be physically implemented by a single component and an operation of a single functional unit may be physically implemented by a plurality of components. Concerning the processing procedures described above in the embodiments, the orders of steps may be changed unless a contradiction arises. For the sake of convenience for describing the processing, the base station **10** and the terminal **20** have been described with the use of the functional block diagrams, but these apparatuses may be implemented by hardware, software, or a combination thereof. Each of software functioning with a processor of the base station **10** according to the embodiment of the present invention and software functioning with a processor of the terminal **20** according to the embodiment of the present invention may be stored in a random access memory (RAM), a flash memory, a read-only memory (ROM), an EPROM, an EEPROM, a register, a hard disk (HDD), a removable disk, a CD-ROM, a database, a server, or any suitable recording media.

(93) Also, the notification of information is not limited to the aspect or embodiment described in the present disclosure, but may be performed by other methods. For example, the notification of information may be performed by physical layer signaling (for example, DCI (Downlink Control Information), UCI (Uplink Control Information)), higher layer signaling (for example, RRC (Radio Resource Control) signaling, MAC (Medium Access Control) signaling, broadcast information (an MIB (Master Information Block) and an SIB (System Information Block)), other signals, or combinations thereof. The RRC signaling may be also be referred to as an RRC message and may be, for example, an RRC connection setup message, an RRC connection reconfiguration message, or the like.

(94) Each aspect and embodiment described in the present disclosure may be applied to at least one of a system that uses a suitable system such as LTE (Long Term Evolution), LTE-A (LTE-Advanced), SUPER 3G, IMT-Advanced, 4G (4th generation mobile communication system), 5G (5th generation mobile communication system), FRA (Future Radio Access), NR (New Radio), W-CDMA (registered trademark), GSM (registered trademark), CDMA2000, UMB (Ultra Mobile Broadband), IEEE 802.11 (Wi-Fi (registered trademark)), IEEE 802.16 (WiMAX (registered trademark)), IEEE 802.20, UWB (Ultra-WideBand), or Bluetooth (registered trademark), and a next-generation system expanded on the basis thereof. Also a plurality of systems may be combined and applied (for example, a combination of at least one of LTE and LTE-A with 5G, and the like).

(95) In the operation procedures, sequences, flowcharts, and the like according to each aspect and embodiment described in the present disclosure, the orders of steps may be changed unless a contradiction arises. For example, in the methods described in the present disclosure, elements of various steps are illustrated by using an exemplary order and the methods are not limited to the

specific orders presented.

(96) The specific operations performed by the base station **10** described in the present disclosure may in some cases be performed by an upper node. It is clear that, in a network that includes one or more network nodes including the base station **10**, various operations performed for communication with the terminal **20** can be performed by at least one of the base station **10** and another network node other than the base station **10** (for example, a MME, a S-GW, or the like may be mentioned, but not limited thereto). In the above, the description has been made for the case where another network node other than the base station **10** is a single node as an example. But the another network node may be a combination of a plurality of other network nodes (for example, a MME and a S-GW).

(97) Information, signals, or the like described in the present disclosure may be output from a higher layer (or a lower layer) to a lower layer (or a higher layer). Information, signals, or the like described in the present disclosure may be input and output via a plurality of network nodes.

(98) Information or the like that has been input or output may be stored at a predetermined place (for example, a memory) and may be managed with the use of a management table. Information or the like that is input or output can be overwritten, updated, or appended. Information or the like that has been output may be deleted. Information or the like that has been input may be transmitted to another apparatus.

(99) In the present disclosure, determination may be made with the use of a value expressed by one bit (0 or 1), may be made with the use of a Boolean value (true or false), and may be made through a comparison of numerical values (for example, a comparison with a predetermined value).

(100) Regardless of whether software is referred to as software, firmware, middleware, microcode, a hardware description language, or another name, software should be interpreted broadly to mean instructions, instruction sets, codes, code segments, program codes, a program, a sub-program, a software module, an application, a software application, a software package, a routine, a subroutine, an object, an executable file, an execution thread, a procedure, a function, and the like.

(101) Software, instructions, information, or the like may be transmitted and received through transmission media. For example, in a case where software is transmitted from a website, a server or another remote source through at least one of wired technology (such as a coaxial cable, an optical-fiber cable, a twisted pair, or a digital subscriber line (DSL)) and radio technology (such as infrared or microwaves), at least one of the wired technology and the radio technology is included in the definition of a transmission medium.

(102) Information, signals, and the like described in the present disclosure may be expressed with the use of any one of various different technologies. For example, data, instructions, commands, information, signals, bits, symbols, chips, and the like mentioned herein throughout the above explanation may be expressed by voltages, currents, electromagnetic waves, magnetic fields or magnetic particles, optical fields or photons, or any combinations thereof.

(103) The terms described in the present disclosure and the terms necessary for understanding the present disclosure may be replaced with terms having the same or similar meanings. For example, at least one of a channel and a symbol may be a signal (signaling). A signal may be a message. A component carrier (CC) may be referred to as a carrier frequency, a cell, a frequency carrier, or the like.

(104) The terms “system” and “network” used in the present disclosure are used interchangeably.

(105) Information, parameters, and the like described in the present disclosure may be expressed by absolute values, may be expressed by relative values with respect to predetermined values, and may be expressed by corresponding different information. For example, radio resources may be indicated by indexes.

(106) The above-described names used for the parameters are not restrictive in any respect. In addition, formulas or the like using these parameters may be different from those explicitly disclosed in the present disclosure. Various channels (for example, a PUCCH, a PDCCH, and the

like) and information elements can be identified by any suitable names, and therefore, various names given to these various channels and information elements are not restrictive in any respect. (107) In the present disclosure, terms such as “base station (BS)”, “radio base station”, “base station apparatus”, “fixed station”, “NodeB”, “eNodeB (eNB)”, “gNodeB (gNB)”, “access point”, “transmission point”, “reception point”, “transmission/reception point”, “cell”, “sector”, “cell group”, “carrier”, “component carrier”, and the like may be used interchangeably. A base station may be referred to as a macro-cell, a small cell, a femtocell, a pico-cell, or the like.

(108) A base station can accommodate one or a plurality of (for example, three) cells (that may be called sectors). In a case where a base station accommodates a plurality of cells, the whole coverage area of the base station can be divided into a plurality of smaller areas. For each smaller area, a base station subsystem (for example, an indoor miniature base station RRH (Remote Radio Head)) can provide a communication service. The term “cell” or “sector” denotes all or a part of the coverage area of at least one of a base station and a base station subsystem that provides communication services in the coverage.

(109) In the present disclosure, terms such as “mobile station (MS)”, “user terminal”, “user equipment (UE)”, and “terminal” may be used interchangeably.

(110) By the person skilled in the art, a mobile station may be referred to as any one of a subscriber station, a mobile unit, a subscriber unit, a wireless unit, a remote unit, a mobile device, a wireless device, a wireless communication device, a remote device, a mobile subscriber station, an access terminal, a mobile terminal, a wireless terminal, a remote terminal, a handset, a user agent, a mobile client, a client, and other suitable terms.

(111) At least one of a base station and a mobile station may be referred to as a transmitting apparatus, a receiving apparatus, a communication apparatus, or the like. At least one of a base station and a mobile station may be an apparatus mounted on a mobile body, or may be a mobile body itself, or the like. A mobile body may be a transporting device (e.g., a vehicle, an airplane, and the like), an unmanned mobile (e.g., a drone, an automated vehicle, and the like), or a robot (of a manned or unmanned type). It is noted that at least one of a base station and a mobile station includes an apparatus that does not necessarily move during a communication operation. For example, at least one of a base station and a mobile station may be an IoT (Internet of Thing) device such as a sensor.

(112) In addition, a base station according to the present disclosure may be read as a user terminal. For example, each aspect or embodiment of the present disclosure may be applied to a configuration in which communication between a base station and a user terminal is replaced by communication between a plurality of terminals **20** (that may be called D2D (Device-to-Device), V2X (Vehicle-to-Everything), or the like). In this case, a terminal **20** may have above-described functions of the base station **10**. In this regard, a word such as “up” or “down” may be read as a word corresponding to communication between terminals (for example, “side”). For example, an uplink channel, a downlink channel, or the like may be read as a side channel.

(113) Similarly, a user terminal according to the present disclosure may be replaced with a base station. In this case, a base station may have above-described functions of the user terminal.

(114) The term “determine” used herein may mean various operations. For example, judging, calculating, computing, processing, deriving, investigating, looking up, searching, inquiring (for example, looking up a table, a database, or another data structure), ascertaining, or the like may be deemed as making determination. Also, receiving (for example, receiving information), transmitting (for example, transmitting information), inputting, outputting, or accessing (for example, accessing data in a memory), or the like may be deemed as making determination. Also, resolving, selecting, choosing, establishing, comparing, or the like may be deemed as making determination. That is, doing a certain operation may be deemed as making determination. “To determine” may be read as “to assume”, “to expect”, “to consider”, or the like.

(115) Each of the terms “connected” and “coupled” and any variations thereof mean any

connection or coupling among two or more elements directly or indirectly and can mean that one or a plurality of intermediate elements are inserted among two or more elements that are “connected” or “coupled” together. Coupling or connecting among elements may be physical one, may be logical one, and may be a combination thereof. For example, “connecting” may be read as “accessing”. In a case where the terms “connected” and “coupled” and any variations thereof are used in the present disclosure, it may be considered that two elements are “connected” or “coupled” together with the use of at least one type of a medium from among one or a plurality of wires, cables, and printed conductive traces, and in addition, as some non-limiting and non-inclusive examples, it may be considered that two elements are “connected” or “coupled” together with the use of electromagnetic energy such as electromagnetic energy having a wavelength of the radio frequency range, the microwave range, or the light range (including both of the visible light range and the invisible light range).

(116) A reference signal can be abbreviated as an RS (Reference Signal). A reference signal may be referred to as a pilot depending on an applied standard.

(117) A term “based on” used in the present disclosure does not mean “based on only” unless otherwise specifically noted. In other words, a term “base on” means both “based on only” and “based on at least”.

(118) Any references to elements denoted by a name including terms such as “first” or “second” used in the present disclosure do not generally limit the amount or the order of these elements. These terms can be used in the present disclosure as a convenient method for distinguishing one or a plurality of elements. Therefore, references to first and second elements do not mean that only the two elements can be employed or that the first element should be, in some way, prior to the second element.

(119) “Means” in each of the above apparatuses may be replaced with “unit”, “circuit”, “device”, or the like.

(120) In a case where any one of “include”, “including”, and variations thereof is used in the present disclosure, each of these terms is intended to be inclusive in the same way as the term “comprising”. Further, the term “or” used in the present disclosure is intended to be not exclusive-or.

(121) A radio frame may include, in terms of time domain, one or a plurality of frames. Each of one or a plurality of frames may be referred to as a subframe in terms of time domain. A subframe may include, in terms of time domain, one or a plurality of slots. A subframe may have a fixed time length (e.g., 1 ms) independent of Numerology.

(122) Numerology may be a communication parameter that is applied to at least one of transmission and reception of a signal or a channel. Numerology may mean, for example, at least one of a subcarrier spacing (SCS), a bandwidth, a symbol length, a cyclic prefix length, a transmission time interval (TTI), the number of symbols per TTI, a radio frame configuration, a specific filtering processing performed by a transceiver in frequency domain, a specific windowing processing performed by a transceiver in time domain, and the like.

(123) A slot may include, in terms of time domain, one or a plurality of symbols (OFDM (Orthogonal Frequency Division Multiplexing) symbols, SC-FDMA (Single Carrier Frequency Division Multiplexing) symbols) symbols, or the like). A slot may be a time unit based on Numerology.

(124) A slot may include a plurality of minislots. Each minislot may include one or a plurality of symbols in terms of the time domain. A minislot may also be referred to as a subslot. A minislot may include fewer symbols than a slot. A PDSCH (or PUSCH) transmitted at a time unit greater than a minislot may be referred to as a PDSCH (or PUSCH) mapping type A. A PDSCH (or PUSCH) transmitted using minislots may be referred to as a PDSCH (or PUSCH) mapping type B.

(125) Each of a radio frame, a subframe, a slot, a minislot, and a symbol means a time unit configured to transmit a signal. Each of a radio frame, a subframe, a slot, a minislot, and a symbol

may be referred to as other names respectively corresponding thereto.

(126) For example, one subframe may be referred to as a transmission time interval (TTI), a plurality of consecutive subframes may be referred to as a TTI, and one slot or one minislot may be referred to as a TTI. That is, at least one of a subframe and a TTI may be a subframe (1 ms) according to the existing LTE, may have a period shorter than 1 ms (e.g., 1 to 13 symbols), and may have a period longer than 1 ms. Instead of subframes, units expressing a TTI may be referred to as slots, minislots, or the like.

(127) A TTI means, for example, a minimum time unit of scheduling in radio communication. For example, in an LTE system, a base station performs scheduling for each terminal **20** to assign, in TTI units, radio resources (such as frequency bandwidths, transmission power, and the like that can be used by each terminal **20**). However, the definition of a TTI is not limited thereto.

(128) A TTI may be a transmission time unit for channel-coded data packets (transport blocks), code blocks, code words, or the like, and may be a unit of processing such as scheduling, link adaptation, or the like. When a TTI is given, an actual time interval (e.g., the number of symbols) to which transport blocks, code blocks, code words, or the like are mapped may be shorter than the given TTI.

(129) In a case where one slot or one minislot is referred to as a TTI, one or a plurality of TTIs (i.e., one or a plurality of slots or one or a plurality of minislots) may be a minimum time unit of scheduling. The number of slots (the number of minislots) included in the minimum time unit of scheduling may be controlled.

(130) A TTI having a time length of 1 ms may be referred to as an ordinary TTI (a TTI according to LTE Rel. 8-12), a normal TTI, a long TTI, an ordinary subframe, a normal subframe, a long subframe, a slot, or the like. A TTI shorter than an ordinary TTI may be referred to as a shortened TTI, a short TTI, a partial or fractional TTI, a shortened subframe, a short subframe, a minislot, a subslot, a slot, or the like.

(131) Note that a long TTI (for example, normal TTI, subframe, and the like) may be read as TTI having a time length exceeding 1 ms, and a short TTI (for example, shortened TTI) may be read as a TTI having a TTI length less than the TTI length of the long TTI and equal to or more than 1 ms.

(132) A resource block (RB) is a resource assignment unit in terms of time domain and frequency domain and may include one or a plurality of consecutive subcarriers in terms of frequency domain. The number of subcarriers included in an RB may be the same regardless of Numerology, and, for example, may be 12. The number of subcarriers included in a RB may be determined based on Numerology.

(133) In terms of time domain, an RB may include one or a plurality of symbols, and may have a length of 1 minislot, 1 subframe, or 1 TTI. Each of 1 TTI, 1 subframe, and the like may include one or a plurality of resource blocks.

(134) One or a plurality of RBs may be referred to as physical resource blocks (PRBs: Physical RBs), a subcarrier group (SCG: Sub-Carrier Group), a resource element group (REG: Resource Element Group), a PRB pair, an RB pair, or the like.

(135) A resource block may include one or a plurality of resource elements (RE: Resource Elements). For example, 1 RE may be a radio resource area of 1 subcarrier and 1 symbol.

(136) A bandwidth part (BWP) (which may be called a partial bandwidth or the like) may mean a subset of consecutive common RBs (common resource blocks) for Numerology, in any given carrier. A common RB may be identified by a RB index with respect to a common reference point in the carrier. PRBs may be defined by a BWP and may be numbered in the BWP.

(137) A BWP may include a BWP (UL BWP) for UL and a BWP (DL BWP) for DL. For a UE, one or a plurality of BWPs may be set in 1 carrier.

(138) At least one of BWPs that have been set may be active, and a UE need not assume sending or receiving a predetermined signal or channel outside the active BWP. A “cell”, a “carrier” or the like in the present disclosure may be read as a “BWP”.

(139) The above-described structures of radio frames, subframes, slots, minislots, symbols, and the like are merely examples. For example, the number of subframes included in a radio frame, the number of slots included in a subframe or a radio frame, the number of minislots included in a slot, the number of symbols and the number of RBs included in a slot or a minislot, the number of subcarriers included in an RB, the number of symbols included in a TTI, a symbol length, a cyclic prefix (CP) length, and the like can be variously changed.

(140) Throughout the present disclosure, in a case where an article such as “a”, “an”, or “the” in English is added through a translation, the present disclosure may include a case where a noun following such article is of a plural forms.

(141) Throughout the present disclosure, an expression that “A and B are different” may mean that “A and B are different from each other”. Also this term may mean that “each of A and B is different from C”. Terms such as “separate” and “coupled” may also be interpreted in a manner similar to “different”.

(142) Each aspect or embodiment described in the present disclosure may be solely used, may be used in combination with another embodiment, and may be used in a manner of being switched with another embodiment upon implementation. Notification of predetermined information (for example, notification of “being x”) may be implemented not only explicitly but also implicitly (for example, by not notifying predetermined information).

(143) In the present disclosure, the OFDM is an example of a multi-carrier signal.

(144) Although the present disclosure has been described above, it will be understood by those skilled in the art that the present disclosure is not limited to the embodiment described in the present disclosure. Modifications and changes of the present disclosure may be possible without departing from the subject matter and the scope of the present disclosure defined by claims. Therefore, the descriptions of the present disclosure are for illustrative purposes only, and are not intended to be limiting the present disclosure in any way.

REFERENCE SIGNS LIST

(145) **10** base station **110** transmitting unit **120** receiving unit **130** configuring unit **140** control unit **20** terminal **210** transmitting unit **220** receiving unit **230** configuring unit **240** control unit **30** core network **1001** processor **1002** storage device **1003** auxiliary storage device **1004** communication apparatus **1005** input device **1006** output device

Claims

1. A terminal comprising: a communication unit configured to transmit and receive a multi-carrier signal in a frequency band equal to or higher than a predetermined frequency; and a control unit configured to determine a first frame structure of the multi-carrier signal, wherein in the first frame structure, at least one of a subcarrier spacing and a number of resource blocks in a maximum channel bandwidth, a CP (cyclic prefix) ratio, and a time unit or a frequency unit of scheduling is different from a second frame structure of the multi-carrier signal in a frequency band equal to or lower than the predetermined frequency, and wherein in the first frame structure, the time unit or the frequency unit of the scheduling is larger than that in the second frame structure.
2. The terminal according to claim 1, wherein in the first frame structure, the subcarrier spacing is larger than that in the second frame structure, and the number of resource blocks in the maximum channel bandwidth is smaller than that in the second frame structure.
3. The terminal according to claim 1, wherein in the first frame structure, a ratio of a CP length to a symbol length is smaller than that in the second frame structure.
4. The terminal according to claim 3, wherein the first frame structure has a same subcarrier spacing as the second frame structure, and includes more symbols in a given time length than the second frame structure.
5. A communication method causing a terminal to execute: transmitting and receiving a multi-

carrier signal in a frequency band equal to or higher than a predetermined frequency; and determining a first frame structure of the multi-carrier signal, wherein in the first frame structure, at least one of a subcarrier spacing and a number of resource blocks in a maximum channel bandwidth, a CP (cyclic prefix) ratio, and a time unit or a frequency unit of scheduling is different from a second frame structure of the multi-carrier signal in a frequency band equal to or lower than the predetermined frequency, and wherein in the first frame structure, the time unit or the frequency unit of the scheduling is larger than that in the second frame structure.
